Koneru Lakshmaiah Education Foundation
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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2 -
Academic Year	I-III Semester (2025 26)
Faculty Name	Swati
Student Name	Kandula Poojitha Rani
Roll Number	2520030460
Branch	CSE
Course Outcome	

Case Study Topic: Segment Tree for Sales Range Analytics

Work Done Summary: This case study implements a Segment Tree in Java for sales range analytics in the RetailMind – Smart Retail Inventory & Sales Analytics System. The program stores and manages retail sales data using a Segment Tree structure to support efficient sales analysis. Java classes and methods were developed to build the tree, perform range sum queries, and update sales records dynamically. The system helps analyze sales over different time periods, compare performance, and retrieve data quickly without scanning the entire dataset. Segment Tree operations improve processing speed with **O(log n)** query and update complexity, making the system suitable for real-time retail sales monitoring and decision-making.

Implementation Details:

1. Created a **Segment Tree** to store and manage retail sales data.
2. Built the tree using **Java methods** for efficient data organization.
3. Implemented **range query operations** to calculate sales within a selected period.
4. Added **update operations** to modify sales data dynamically.
5. Used the system in **RetailMind** to analyze and monitor sales performance.
6. Improved efficiency with **O(log n)** query and update operations.

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Result:Screenshots/Output:

Output

```
Sales from Day 1 to Day 3: 4500
Updated Sales from Day 1 to Day 3: 5000

=== Code Execution Successful ===
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The Segment Tree based sales analytics system successfully manages and analyzes retail sales data in the RetailMind application. It enables fast range queries and dynamic updates with **O(log n)** time complexity, improving sales monitoring, data retrieval, and decision-making efficiency in retail operations.

GitHub Link: https://github.com/kandulapoojitharani-ai/The-AVL-Tree-based-inventory-system-in-RetailMind/blob/main/ CO-1RetailMind – Smart Retail Inventory & Sales Analytics System

Student Signature	Faculty Signature
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